Ways to Improve the Strategy

After reviewing the backtest results and researching what the industry and academic literature are doing in football prediction and sports betting, here are the concrete improvements that would have the highest impact on the strategy's performance.

1. Prioritize Calibration Over Accuracy

This is probably the single highest-leverage change. A recent study published in Machine Learning with Applications (2025) tested ML models on NBA betting and found that selecting models based on calibration rather than accuracy led to dramatically better returns: +34.69% ROI for the calibration-selected model versus -35.17% for the accuracy-selected model. The reason is straightforward: Kelly criterion only works if the probability estimates are well-calibrated. If the model says 70% but the true probability is 55%, the Kelly formula will overbet, and the bankroll erodes.

Right now, we evaluate Football-LLM on result accuracy (52.3%) and exact score match (29.7%). We don't measure calibration at all. The fix is to add a calibration-aware training objective or, more practically, to apply post-hoc calibration (Platt scaling or isotonic regression) to the model's output probabilities. This requires a held-out calibration set, which could be carved out of the existing training data or collected from league matches.

We should also shift the primary evaluation metric from accuracy to Ranked Probability Score (RPS), which is the standard in the football prediction literature (used in the 2017 Soccer Prediction Challenge). RPS rewards well-distributed probability estimates, not just getting the most likely outcome right. A model with 50% accuracy but excellent calibration will outperform a model with 55% accuracy but poor calibration in a Kelly betting framework.

2. Integrate Elo and Pi-Ratings as Additional Context

The football prediction literature is consistent on this point: Elo-based and pi-rating features are the strongest single predictors of match outcomes. The best result in the 2017 Soccer Prediction Challenge came from CatBoost + pi-ratings (55.82% accuracy, 0.1925 RPS), and subsequent work has confirmed that these rating features outperform raw player statistics in most settings. Our model uses only player-level stats, which capture squad quality but miss the team-level momentum signals that ratings capture (recent form, strength of schedule, home/away performance trajectory).

The practical fix is to add Elo and/or pi-ratings directly into the structured prompt, alongside the existing player statistics. Something like: "Team A current Elo: 1842 (+15 over last 5 matches). Team B current Elo: 1756 (-8 over last 5 matches)." These ratings are publicly available for every national team and most club teams. This gives the model both the granular player-level input and the macro-level team trajectory, which is information that no single data source provides alone.

3. Massively Expand the Training Data

384 training samples is far below what the literature uses. The Open International Soccer Database contains 216,000+ matches across 35 countries and 18 seasons. Even if we restrict to matches where player-level data is available, we could likely construct thousands of training samples from the top 5 European leagues alone. More data would help the model in several ways: better generalization to different match types (group stage vs. knockout, league vs. cup), more robust learning of the relationship between player statistics and match outcomes, and a larger held-out set for calibration analysis.

The data pipeline already supports this. The API-Football scraper collects player stats across all major leagues. The prompt generation code can be reused directly. The main effort is in collecting starting XI data for historical league matches (to know which players actually played) and matching them to the player season stats.

4. Ensemble with Traditional ML Models

The academic literature is clear that ensemble methods outperform any single model for football prediction. Modern prediction platforms combine Poisson distribution models (for goal probability estimation), Elo/pi-ratings (for team strength), XGBoost or CatBoost (for match outcome classification), and sometimes LSTM networks (for form and momentum). The most reliable platforms reportedly achieve 70%+ accuracy by combining multiple approaches.

Our LLM does not need to replace these approaches. Instead, it should be one voice in an ensemble. A practical architecture would be: run XGBoost with Elo + player features as a tabular model, run the LLM for prediction + reasoning, then combine their probability outputs using a simple weighted average or a learned meta-model. The LLM's unique contribution is the reasoning chain (which no tabular model can provide) and its ability to handle compositional roster data without feature engineering. The tabular model's contribution is better-calibrated probabilities from a larger training set. Together, they cover each other's weaknesses.

5. Add Real-Time Data Integration

The current model predicts from a static snapshot of player statistics aggregated over the prior 3 seasons. It has no awareness of recent form (last 5 matches), injuries announced in the days before the match, tactical changes, or weather conditions. Professional betting models and bookmakers continuously update their estimates as new information arrives.

Two practical additions: first, include a "recent form" section in the prompt that summarizes each team's last 5-10 match results and key statistics from those matches. This is cheap to compute from existing data and gives the model access to momentum signals. Second, integrate a news/injury feed (even a simple one scraped from a football news API) that flags key absences and lineup changes. This information could either be added to the prompt directly or used as a post-prediction filter in the reasoning audit step.

6. Increase Training Sequence Length and Add Expected Goals (xG)

A technical improvement: the current training uses max_length=768 tokens, which forced us to compress team profiles heavily. Increasing to 2048 tokens (feasible on an A100 or even a T4 with gradient checkpointing and smaller batch sizes) would allow richer prompts containing individual player stat lines for the starting XI, historical head-to-head details, and recent form data. The initial version of the project actually had verbose prompts that were truncated at 768 tokens, causing the model to never see the assistant response during training. The compact prompts fixed this, but they sacrificed a lot of information.

Additionally, expected goals (xG) data has become one of the strongest predictors in modern football analytics. Post-match xG analysis achieves about 65.6% accuracy in predicting match outcomes with an RPS of 0.148. While we can't use post-match xG for pre-match prediction, pre-match xG estimates derived from shot location models (publicly available from sources like FBref and Understat) could be incorporated as features. These capture shot quality and chance creation in a way that raw goal counts do not.

7. Optimize for the Betting Market, Not Just Prediction Accuracy

The current backtest simulates flat betting at assumed average odds. A more sophisticated approach would integrate actual historical bookmaker odds and optimize the strategy specifically for expected value (EV), not accuracy. The Prophet Arena benchmark found that even frontier LLMs struggle to be consistently profitable against prediction markets, because being "right" 52% of the time only matters if the odds compensate for the 48% you're wrong.

Concretely, this means: instead of training the model to maximize prediction accuracy, consider a loss function or evaluation metric that weights predictions by their potential Kelly value. A correct prediction on a 3.0 odds underdog is worth more than a correct prediction on a 1.3 odds favorite, and the training data should reflect this. Additionally, incorporating bookmaker odds as a feature in the prompt (so the model can see what the market thinks) could help it identify divergences rather than trying to predict match outcomes in isolation.